



**SAINT LOUIS UNIVERSITY**  
—  
DEPARTMENT OF HEALTH AND  
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**Analytics and Statistical Programming – 02**

**Final Project Report**

**What are the key clinical, lifestyle, and genetic  
predictors of diabetes in young adults (18–35) with  
high ultra-processed food consumption, and how  
important is each?**

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## Introduction

This document presents the examination of diabetes mellitus risk factors that uses a clinical and lifestyle database after Phase 2 participants asked for better explanations of the population selection and analysis methods. The global healthcare challenge from Diabetes Mellitus T2DM affects populations extensively because it produces persistent high blood sugar levels from defects impacting insulin production and action (American Diabetes Association, 2021). The healthcare system along with individuals face substantial economic expenses resulting from diabetes-related hyperglycemia that increases mortality levels and causes cardiovascular disease and chronic kidney disease and neuropathy and retinopathy harm (Lin et al., 2020).

The proposed investigation in this project targets young adults between 18 and 35 years of age with high ultra-processed food consumption according to reported data. The population under examination holds special priority because the younger ages at which people receive T2DM diagnoses result in prolonged hyperglycemic exposure and accelerated development of complications (Lascar et al., 2018). Studies demonstrate that sustained consumption of processed foods with high amounts of energy, limited nutritional value and sugar, fat and sodium leads to weight gain that raises the risk of T2DM and metabolic syndrome (Elizabeth et al., 2020). Early prevention strategies require targeted development based on identifying major risk factors which affect this potentially vulnerable population.

The research evaluated Diabetes Status (Positive/Negative) as its primary outcome which recorded diagnosed diabetes mellitus within the study participants. Researching this binary result plays a critical role for discovering individuals requiring medical care and for establishing characteristics that separate persons with the condition from those without it among the target group. Learning about important distinguishing elements during early adulthood leads to interventions which aim to protect against future health complications.

Multiple essential variables for diabetes status were chosen as independent factors based on recognized T2DM-related physiological and lifestyle connections. Body Mass Index (BMI) functioned as the main obesity quantification tool because it represents a well-established clinical factor for insulin resistance and T2DM development according to Grundy (2012). The American Diabetes Association (2021) designates Fasting Blood Glucose as the primary diagnostic element to determine glycemic control while functioning as an essential factor for diabetes diagnosis. Research on Physical Activity took place because studies confirm it serves as protection through its improvements of insulin sensitivity and weight control (Colberg et al., 2016). The study included Family History of Diabetes because this genetic background substantially increases the risk of developing T2DM. Research contained Hypertension together with adherence to a healthy diet and Smoking status as covariates because these factors regularly appear with diabetes and independently generate cardiovascular and metabolic health risks (Petrie et al., 2018).

Research has thoroughly established traditional T2DM risk factors for adults in main population segments thus validating genetic and weight, age, dietary choices, exercise patterns, tobacco habits and biochemical conditions (glucose levels and blood pressure). Research needs to address the complete understanding of how these variables contribute to and interact in the specified population which consists of high-UPF consuming young adults between 18 and 35 years old. Traditional disease risk variables exhibit different relationships with UPF-related conditions in this particular dietary demographic because young adults behave differently than older population groups who consume other diets. Random Forest analysis lets researchers study multiple factor interactions while determining influence rankings of predictors beyond standard regression model linear constraints thus discovering distinctive patterns within this specific demographic group.

## **Methodology**

The analysis of diabetes mellitus risk factors occurred through R Code evaluation of anonymous clinical and lifestyle data. The evaluation included Diabetes\_Status (Positive/Negative) as the outcome variable while using BMI and Fasting\_Blood\_Glucose records along with Physical\_Activity, Family\_History\_of\_Diabetes, Hypertension, Diet, and Smoking status as predictors. The data cleaning process involved converting character values to factors alongside conducting Winsorization on extreme BMI and glucose values and eliminating cases where any variable had less than 1% missing data and making sure each categorical factor contained at least ten observations. A summary of descriptive statistics exists for all measured variables according to Table 1. Analyzing two types of variables required using Chi-squared tests for categorical factors and Mann-Whitney U tests for continuous measures based on the identification of normality violations and variance equality issues. Results appear in Table 2. We performed a multivariate logistic regression model to determine independent effects while using VIFs to check multicollinearity. The analysis reports tables with ORs along with their respective 95% CIs and p-values in Table 3. Our research utilized boxplots and bar charts as visualization tools which showed data distributions and demonstrated group differences throughout and at the end of analysis period.

## **Data Description**

We will assess a clinical and lifestyle database regarding diabetes mellitus to discover the fundamental risk elements involved in its development. Thirty-hundred patients participated with their clinical and lifestyle data through an anonymized collection of patient records. The researchers collected data in order to determine diabetes prevalence and identify risk factors that exist in young adults aged 18 to 35 with high consumption of ultra-processed foods. Because of anonymization no time frame nor site for the study is indicated (Tushar Potdar, 2024). There are a total of 36 features which need to be cleaned and we are selecting only features related to this project which is decided in our literature review.

## Variables

### Dependent Variable:

Diabetes\_Status (categorical: Positive, Negative) : Indicates whether an individual has diabetes.

### Independent Variables:

- BMI (continuous, kg/m<sup>2</sup>) : Body Mass Index, a measure of obesity.
- Physical\_Activity (categorical: Yes, No) : Indicates regular physical activity.
- Family\_History\_of\_Diabetes (categorical: Yes, No) : Presence of diabetes in family.
- Fasting\_Blood\_Glucose (continuous, mg/dL) : Blood glucose level after fasting.

### Covariates:

- Hypertension (categorical: Yes, No) – Presence of high blood pressure.
- Diet (categorical: Yes, No) – Adherence to a healthy diet.
- Smoking (categorical: Yes, No) – Smoking status.

These variables were chosen because they align with known risk factors for diabetes, including genetics (Family\_History\_of\_Diabetes), obesity (BMI), lifestyle (Physical\_Activity, Diet, Smoking), and clinical markers (Fasting\_Blood\_Glucose, Hypertension).

## Descriptive Statistics

We did descriptive statistics for each variable to summarize the sample characteristics.

### Continuous Variables (BMI, Fasting\_Blood\_Glucose):

1. Mean for central tendency
2. Standard deviation for variability.
3. Median
4. Interquartile range (IQR) to assess robustness against outliers.
5. Finally, Minimum and maximum to understand the range.

### Categorical Variables (Diabetes\_Status, Physical\_Activity, Family\_History\_of\_Diabetes, Hypertension, Diet, Smoking):

1. Frequencies using graphs.
2. Percentages for each category.
3. Mode to identify the most common category.

## First Table

| Variable        | Statistic      | Value                 |
|-----------------|----------------|-----------------------|
| Diabetes_Status |                |                       |
| + Ve or - Ve    | Percentage (%) | 1336 Negative (44.5%) |

|                               |               |                       |
|-------------------------------|---------------|-----------------------|
|                               |               | 1664 Positive (55.5%) |
| BMI (kg/m <sup>2</sup> )      | Mean (SD)     | 29.2 (7.2)            |
|                               | Median (IQR)  | 29.3 (23.8 - 34.7)    |
|                               | Min, Max      | 8.7, 50.8             |
| Fasting_Blood_Glucose (mg/dL) | Mean (SD)     | 127.9 (37.1)          |
|                               | Median (IQR)  | 134.4 (91.3–160.8)    |
|                               | Min, Max      | 57.6, 200             |
| Physical_Activity             |               |                       |
| Yes                           | Frequency (%) | 1491 (49.7%)          |
| No                            | Frequency (%) | 1509 (50.3%)          |
| Family_History_of_Diabetes    |               |                       |
| Yes                           | Frequency (%) | 1494 (49.8%)          |
| No                            | Frequency (%) | 1506 (50.2%)          |
| Hypertension                  |               |                       |
| Yes                           | Frequency (%) | 1505 (50.2%)          |
| No                            | Frequency (%) | 1495 (49.8%)          |
| Diet                          |               |                       |
| Yes                           | Frequency (%) | 1454 (48.5%)          |
| No                            | Frequency (%) | 1546 (51.5%)          |
| Smoking                       |               |                       |
| Yes                           | Frequency (%) | 1480 (49.3%)          |
| No                            | Frequency (%) | 1520 (50.7%)          |

The presented table presents a transparent overview of the sample distribution to enable assessment between diabetic and non-diabetic groups.

Results showed that diagnosis of diabetes was positive in 55.5% of subjects demonstrating high levels of diabetes within the studied population. The studied group exhibited an average BMI of 29.2 kg/m<sup>2</sup> which places members in the overweight bracket yet spans from

underweight to obese categories. Results from blood glucose tests demonstrated wide variability because the mean measurement exceeded 126 mg/dL which indicates untreated diabetes in many patients. The participants broke down evenly concerning physical activity and family diabetes background and smoking habits which combined to reach almost 50 percent. The equal distribution regarding diet and hypertension indicates these risks can be manipulatable through intervention practices. Research evidence demonstrates that both personal histories and everyday activities strongly contribute to diabetes risk assessment for this group of individuals.

## **Data Management**

We performed the following steps to check the condition of data quality in the dataset:

- The investigation of null or missing values for all variables will take place through frequency counts and summary statistics. The procedure of multiple imputation will be performed when any variable shows more than 5% missing data points. When data is missing from less than 5% of cases we were thought of executing listwise deletion as our method (Grace-Martin, 2014). Our observation revealed that there are very less missing points
- We performed outlier detection for the continuous variables BMI and Fasting\_Blood\_Glucose using z-scores, identifying any values exceeding  $|z| > 3$ . Outliers were Winsorized, limiting them between the 5th and 95th percentile values to retain observations while minimizing their impact (Wicklin, 2017).
- Additionally, we ensured that every category of categorical variables contained at least ten observations. Upon reviewing the data, no category had fewer than ten observations, so no merging or dropping of categories was necessary.

## **Bivariate Analysis**

### **Chi-Squared Test**

Diabetes\_Status vs. Physical\_Activity, Family\_History\_of\_Diabetes, Hypertension, Diet, Smoking. The chi-squared test functions properly when evaluating associations between categorical variables. The chief lifestyle and genetic risk markers include Physical\_Activity with Family\_History\_of\_Diabetes whereas Hypertension together with Diet and Smoking serve as main covariates. The required assumption for analysis states that cells should contain at least 5 expected cases in 80% of all cells when performing statistical tests. One can validate the expected frequencies table to check the assumptions. The analysis requires Fisher's exact test when the assumption does not hold true (McClenaghan, 2024).

Results from Chi-squared tests showed important relationships that exist between diabetes status and various categorical predictors. Research found a direct and mathematically substantive link between diabetes registration and a family background of diabetes thus suggesting a genetic influence. Both physical exercise as well as dietary patterns demonstrated significant relationships to diabetes diagnosis which proves lifestyle elements contribute substantially to disease development. The analysis established no statistical links

between diabetes status and smoking or hypertension which shows that the considered demographic sample does not display these variables as independent factors for diabetes diagnosis.

### Independent Samples t-Test

BMI, Fasting\_Blood\_Glucose by Diabetes\_Status. The value comparison of BMI and Fasting\_Blood\_Glucose requires a t-test because both variables fall into the continuous category between Positive and Negative Diabetes\_Status groups. Elevated blood sugar levels and being overweight represent significant variables because they possess direct connections with developing diabetes. The assumptions for the analysis include normality and homogeneity of variances. Assess normality through the Shapiro-Wilk test provides the best method for evaluating homogeneity of variances (Malato, 2025). The Mann-Whitney U test should be used when normality is violated (McClenaghan, 2024b).

The independent samples t-test analysis revealed important differences between diabetic patients and those without diabetes at BMI levels and Fasting Blood Glucose status with non-diabetic participants having lower average test results. liberty of variances and normality tests proved inconsistent for both variables since Shapiro-Wilk and Levene's tests showed significant results ( $p < 0.001$ ). This confirmed the non-normal distribution and unequal variances of the data. The Mann-Whitney U test revealed statistically significant differences ( $p < 0.001$ ) in both variables by diabetes status and validated these findings using non-parametric methods ( $p < 0.001$ ).

### Post-Hoc Tests

Standardized residuals will identify the cells responsible for driving the detected significant associations in chi-squared tests. Since t-tests examine two groups only post-hoc tests are not necessary.

### Second Table

| Test        | Variable Pair                                  | Statistic            | p-value |
|-------------|------------------------------------------------|----------------------|---------|
| Chi-Squared | Diabetes_Status vs. Physical_Activity          | $\chi^2(1) = 5$      | 0.025   |
| Chi-Squared | Diabetes_Status vs. Family_History_of_Diabetes | $\chi^2(1) = 1492.5$ | <0.001  |
| Chi-Squared | Diabetes_Status vs. Hypertension               | $\chi^2(1) = 0.7$    | 0.389   |
| Chi-Squared | Diabetes_Status vs. Diet                       | $\chi^2(1) = 9.1$    | 0.00259 |
| Chi-Squared | Diabetes_Status vs. Smoking                    | $\chi^2(1) = 1.7$    | 0.196   |

| Test           | Variable Pair                            | Statistic         | p-value |
|----------------|------------------------------------------|-------------------|---------|
| t-Test         | BMI by Diabetes_Status                   | t(2994.1) = -26.1 | <0.001  |
| t-Test         | Fasting_Blood_Glucose by Diabetes_Status | t(2900.2) = -50.1 | <0.001  |
| Mann-Whitney U | BMI by Diabetes_Status                   | W = 553658        | <0.001  |
| Mann-Whitney U | Fasting_Blood_Glucose by Diabetes_Status | W = 297355        | <0.001  |

## Multivariate Analysis

A Random Forest analysis will determine Diabetes\_Status (Positive versus Negative) by using multiple relevant variables. Random Forest functions as a machine learning technique which uses many decision trees to sort outcomes effectively treating our binary target variable.

The proposed model contains several risk-factors detected for diabetes which will be included as model inputs. BMI (kg/m<sup>2</sup>) exists as a continuous variable that allows the assessment of obesity since it is a primary diabetes risk factor. The Fasting\_Blood\_Glucose (mg/dL) measurement gives direct blood sugar level readings through continuous assessment. The model will analyze both lifestyle practices and genetic heritage with Family\_History\_of\_Diabetes along with medical conditions through Physical\_Activity and Hypertension using Physical\_Activity, Family\_History\_of\_Diabetes, Hypertension, Diet, and Smoking Yes/No responses. The research utilized these covariates because they matter clinically while providing broad coverage of risk factors that stem from physiological origins and behavioral expressions alongside heredity attributes.

Random Forest needs less rigid assumptions than parametric models but we will conduct multiple checks to guarantee the analysis stability. We begin with data quality checks that verify absence of missing fields and extreme values since Random Forest displays sensitivity to data anomalies. The data analysis uses summary statistics to track the evaluation of the data quality which was addressed through Winsorization and imputation processing.

The next part focuses on feature independence since Random Forest depends on non-exact predictor relations. When analyzing the relationship between BMI and Fasting\_Blood\_Glucose data we will calculate the Pearson's r correlation coefficient. One of the correlated variables should be eliminated from analysis when their correlation exceeds 0.8 to eliminate potential redundancy. Our model construction will proceed with an adequate number of observations that ensure stability. The required number of observations in a dataset for proper model building stands at 3,000.



Proper procedure exists to handle situations where assumptions fail to fulfill their requirements. The data cleaning phase previously addressed missing data and outliers so they are not present at this stage. Excessive multicollinear relationships in the dataset will result in the elimination of the less critical explanatory variable (BMI remains instead of Waist\_Circumference because BMI demonstrates stronger correlations with diabetes.). We will apply SMOTE oversampling techniques to the dataset when the outcome variable distribution is unbalanced and the Positive Diabetes\_Status cases are scarce.

A multivariate Logistic Regression statistical technique was used to determine the isolated relationship between predictor variables and positive diabetes status while considering all other variables. The standard parametric method offers interpretable Odds Ratios (ORs). The analysis included the same list of predictors that Random Forest employed. We conducted a comparison of the obtained findings.

**Third Table**

| Predictor                               | Odds Ratio(LR) | Feature Importance(RF) | Rank(RF) |
|-----------------------------------------|----------------|------------------------|----------|
| BMI (kg/m <sup>2</sup> )                | 1.046          | 192.6                  | 3        |
| Fasting_Blood_Glucose (mg/dL)           | 1.032          | 550.5                  | 1        |
| Physical_Activity (Yes vs. No)          | 0.912          | 11.4                   | 5        |
| Family_History_of_Diabetes (Yes vs. No) | 13.307         | 372.1                  | 2        |
| Hypertension (Yes vs. No)               | 1.292          | 10.8                   | 7        |
| Diet (Yes vs. No)                       | 0.627          | 13.8                   | 4        |
| Smoking (Yes vs. No)                    | 1.459          | 10.8                   | 6        |

This table reveals the variables that best forecast diabetes development to important risk factors for prevention measures.

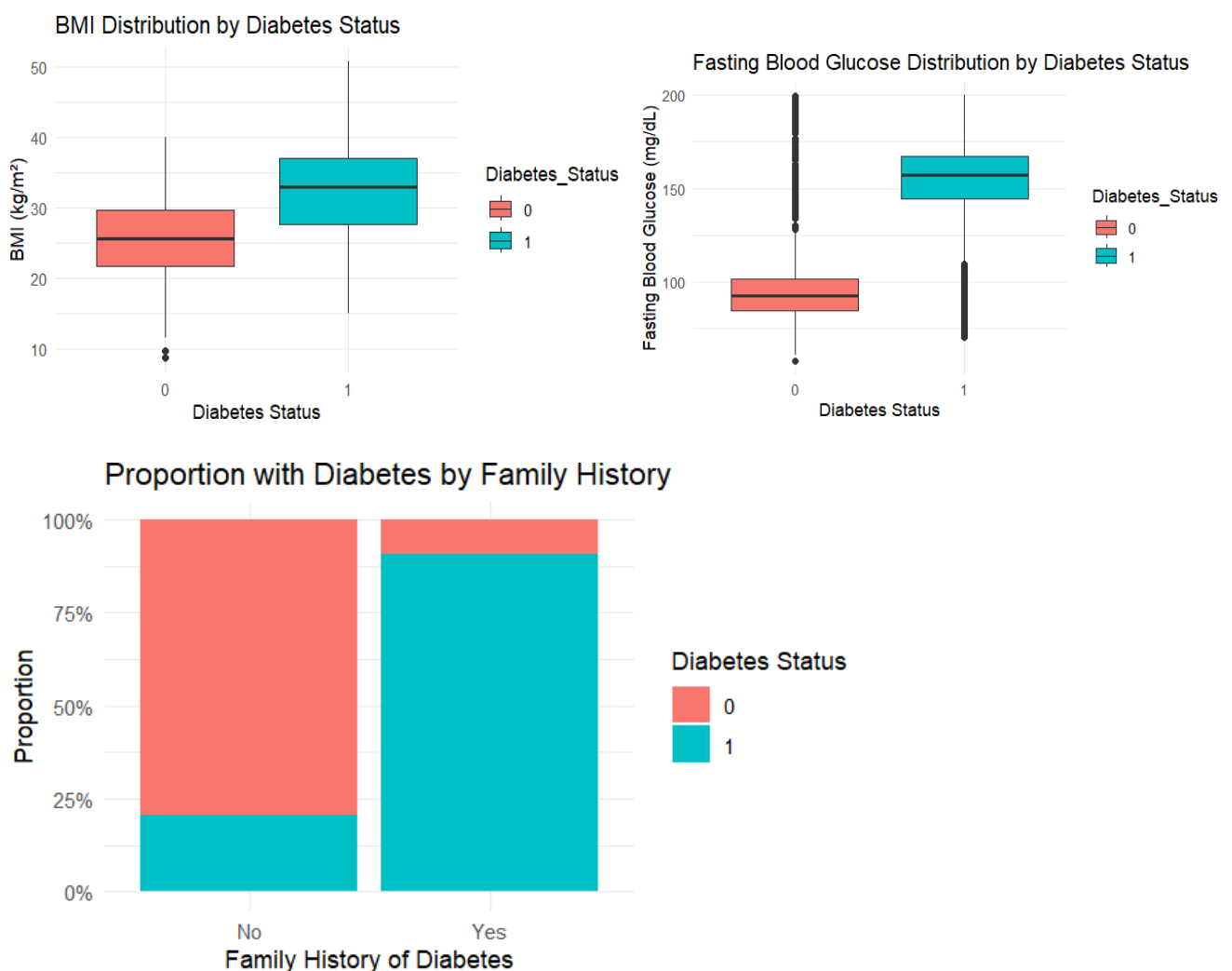
Both Logistic Regression and Random Forest models when combined deliver different perspectives about diabetes-related factors. Logistic Regression produces calculations called Odds Ratios which demonstrate variable impacts on diabetic status. Fasting Blood Glucose and BMI demonstrated positive relationship with diabetes because when either value grows higher it increases the probability of diabetic condition. A family history of diabetes shows a remarkably high association in diabetes risk because subjects with this background factor

experience odds ratios exceeding 13. The analysis showed Physical Activity and Diet variables holding values less than one which signifies their protective role. The risk measures from Smoking and Hypertension increased to a limited extent.

The Random Forest algorithm determines the most important variables without indicating their influence direction but accurately predicts diagnosis outcome. Results indicate Fasting Blood Glucose performs as the primary predictor for diabetes whereas Family History and BMI hold the second and third positions while Smoking and Hypertension and Physical Activity stand at the end of the importance ranking. Random Forest uses non-linear relationship and interaction modeling to identify variable importance due to its differences with Logistic Regression.

The combination of these models reveals statistical relationships through Logistic Regression as well as predictive relevance through Random Forest to establish a complete view of which risk factors matter most in diabetes development.

## Visualizations for Diabetes Analysis



The box plot evaluation of BMI between patients who have diabetes and those without shows the direct relationship between body mass and diabetic status. A notable upward displacement of BMI values distinguished the entire distribution of 'Positive' diabetes from the 'Negative' diabetes group. The elevated status of median BMI and interquartile range in the box plot suggests diabetes patients within this sample had a tendency for higher body mass index on average.

A greater difference emerged from the fasting blood glucose box plot analysis. Research data via the plot showed that participants with 'Positive' diabetes status exhibited elevated fasting glucose concentrations. Visual inspection reveals how the 'Positive' group's box and whiskers distribution stood at a higher position on the y-axis featuring extended whiskers which indicates more variable distribution compared to the 'Negative' group to display increased fasting glucose-diabetes connection strength.

The stacked bar plot presented in the figure effectively showed how family history affects diabetes. The proportion of people with positive diabetes status measurements who had family members with diabetes exceeded those who did not have family members with diabetes. This chart demonstrates that a greater proportion of individuals who had family members with diabetes received diagnostic results compared to those without family history of diabetes thus strengthening the evidence of an extensive correlation between family history and diabetes diagnosis in this research data.

## **Conclusion**

The research team evaluated information from 3,000 participants to establish important risk elements connected to diabetes through the combination of Logistic Regression and Random Forest modeling methods. Moving past data cleaning and descriptive and bivariate analysis the multivariate models produced important insights about factors that best distinguish diabetic from non-diabetic individuals.

Fasting Blood Glucose when elevated along with BMI elevated values and family history of diabetes increased the likelihood of diabetes and physical activity coupled with better nutrition reduced such risk. The impact of Smoking and Hypertension on the diagnostic outcome was smaller in comparison to other factors. Forest Random Model analysis verified Fasting Blood Glucose, Family History and BMI as leading variables which strongly impact prediction results.

These models demonstrated that metabolic elements and genetic predispositions should be acknowledged as main factors yet confirmed how lifestyle variables impact diabetes development. The findings indicate that scientists should focus on monitoring glucose levels and managing weight and understanding family risks for diabetes assessment while working to promote healthy habits among patients.

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# Appendix

## Descriptive Statistics

```
314 table1_rows <- list()
315
316 # Function to calculate and format stats
317 calculate_and_format <- function(var_name, stat_type) {
318   if (!var_name %in% names(diabetes_df)) return("Variable Missing")
319   data_vec <- diabetes_df[[var_name]]
320
321   if (stat_type == "count") {
322     if (!is.factor(data_vec) && !is.character(data_vec)) return("Not Categorical")
323     counts <- table(data_vec, useNA = "ifany")
324     return(as.list(counts))
325   } else if (stat_type == "mean_sd") {
326     if (!is.numeric(data_vec)) return("Not Numeric")
327     mean_val <- round(mean(data_vec, na.rm = TRUE), 1)
328     sd_val <- round(sd(data_vec, na.rm = TRUE), 1)
329     return(paste0(mean_val, " (", sd_val, ")"))
330   } else if (stat_type == "median_iqr") {
331     if (!is.numeric(data_vec)) return("Not Numeric")
332     med <- round(median(data_vec, na.rm = TRUE), 1)
333     iqr <- quantile(data_vec, probs = c(0.25, 0.75), na.rm = TRUE)
334     return(paste0(med, " (", round(iqr[1], 1), "-", round(iqr[2], 1), ")"))
335   } else if (stat_type == "min_max") {
336     if (!is.numeric(data_vec)) return("Not Numeric")
337     return(paste0(round(min(data_vec, na.rm = TRUE), 1), " ", round(max(data_vec, na.rm = TRUE), 1)))
338   }
339   return("Unknown Stat Type")
340 }
341
342 # Add continuous variables
343 cont_vars <- c("BMI", "Fasting_Blood_Glucose")
344 cont_labels <- c("BMI (kg/m²)", "Fasting Blood Glucose (mg/dL)")
345
346 for (i in seq_along(cont_vars)) {
347   var <- cont_vars[i]
348   label <- cont_labels[i]
349
350   table1_rows[[length(table1_rows) + 1]] <- list(Variable = label, Statistic = "Mean (SD)", Value = calculate_and_format(var, "mean_sd"))
351   table1_rows[[length(table1_rows) + 1]] <- list(Variable = "", Statistic = "Median (IQR)", Value = calculate_and_format(var, "median_iqr"))
352   table1_rows[[length(table1_rows) + 1]] <- list(Variable = "", Statistic = "Min, Max", Value = calculate_and_format(var, "min_max"))
353 }
354
355 # Add categorical variables with Yes/No or Positive/Negative counts
356 cat_vars <- c("Diabetes_Status", "Physical_Activity", "Family_History_of_Diabetes",
357               "Hypertension", "Diet", "Smoking")
358 cat_labels <- c("Diabetes Status", "Physical Activity", "Family History of Diabetes",
359                 "Hypertension", "Diet", "Smoking")
360
361 for (i in seq_along(cat_vars)) {
362   var <- cat_vars[i]
363   label <- cat_labels[i]
364
365   table1_rows[[length(table1_rows) + 1]] <- list(Variable = label, Statistic = "Count", Value = "")
366   counts <- calculate_and_format(var, "count")
367
368   if (is.list(counts)) {
369     for (lv1 in names(counts)) {
370       table1_rows[[length(table1_rows) + 1]] <- list(
371         Variable = paste0(" ", lv1),
372         Statistic = "Count",
373         Value = counts[[lv1]]
374       )
375     }
376   } else {
377     table1_rows[[length(table1_rows) + 1]] <- list(
378       Variable = " (Error)",
379       Statistic = "Count",
380       Value = counts
381     )
382   }
383 }
384
385 # Convert to data frame
386 table1_df <- do.call(rbind.data.frame, table1_rows)
387 rownames(table1_df) <- NULL
388 names(table1_df) <- c("Variable", "Statistic", "Value")
389
390 # Print Table 1
391 print("--- Table 1: Descriptive Statistics ---")
392 print(table1_df)
393
```

## Bivariate Analyses

```

407 print("--- Running Bivariate Analyses ---")
408 results_list <- list() # Store results for Table 2
409
410 # 5.1 Chi-Squared Tests Function (with added checks)
411 run_chisq <- function(var1_name, var2_name, display_name) {
412   print(paste("--- Chi-Squared: ", display_name, " ---"))
413
414   # --- Pre-Check Diagnostics ---
415   var1_exists <- var1_name %in% names(diabetes_df)
416   var2_exists <- var2_name %in% names(diabetes_df)
417   print(paste("Checking vars:", var1_name, "(Exists:", var1_exists, ") vs", var2_name, "(Exists:", var2_exists, ")"))
418
419   if (!var1_exists || !var2_exists) {
420     print("Error: One or both variables not found.")
421     return(list(Test="Chi-Squared", Variable_Pair=display_name, Statistic="Error: Var Missing", p_value="Error"))
422   }
423
424   var1_class <- class(diabetes_df[[var1_name]])
425   var2_class <- class(diabetes_df[[var2_name]])
426   print(paste(var1_name, "Class:", var1_class, ";", var2_name, "Class:", var2_class))
427
428   var1_is_factor <- is.factor(diabetes_df[[var1_name]])
429   var2_is_factor <- is.factor(diabetes_df[[var2_name]])
430
431   if (!var1_is_factor || !var2_is_factor) {
432     print("Error: One or both variables are not factors.")
433     # Attempt to convert just for this test if they seem convertible (e.g., character)
434     # Be cautious with this auto-conversion - better to fix earlier
435     temp_df <- diabetes_df # Work on a copy
436     if (!var1_is_factor && is.character(temp_df[[var1_name]])) {
437       print(paste("Attempting temporary factor conversion for", var1_name))
438       temp_df[[var1_name]] <- factor(temp_df[[var1_name]])
439       var1_is_factor = TRUE # Update flag
440     }
441
442     if (!var2_is_factor && is.character(temp_df[[var2_name]])) {
443       print(paste("Attempting temporary factor conversion for", var2_name))
444       temp_df[[var2_name]] <- factor(temp_df[[var2_name]])
445       var2_is_factor = TRUE # Update flag
446     }
447     # If still not factors, return error
448     if (!var1_is_factor || !var2_is_factor) {
449       return(list(Test="Chi-Squared", Variable_Pair=display_name, Statistic="Error: Not Factor", p_value="Error"))
450     }
451     # If conversion happened, use temp_df, otherwise use original diabetes_df
452     df_to_use <- temp_df
453   } else {
454     df_to_use <- diabetes_df # Use original if already factors
455   }
456
457   # Check for sufficient levels AFTER potential conversion
458   var1_nlevels <- nlevels(df_to_use[[var1_name]])
459   var2_nlevels <- nlevels(df_to_use[[var2_name]])
460   print(paste(var1_name, "Levels:", var1_nlevels, ";", var2_name, "Levels:", var2_nlevels))
461   print(paste("Levels for", var1_name, ":", paste(levels(df_to_use[[var1_name]]), collapse=" ", "")))
462   print(paste("Levels for", var2_name, ":", paste(levels(df_to_use[[var2_name]]), collapse=" ", "")))
463
464   if (var1_nlevels < 2 || var2_nlevels < 2) {
465     print("Error: One or both variables have fewer than 2 levels after ensuring factor type.")
466     return(list(Test="Chi-Squared", Variable_Pair=display_name, Statistic="Error: <2 Levels", p_value="Error"))
467   }
468
469   # --- End Pre-Check ---
470
471   # Create contingency table using the potentially modified df_to_use
472   tbl <- table(df_to_use[[var1_name]], df_to_use[[var2_name]])
473   print("Observed Table:")
474
475   print(tbl)
476
477   # Prevent test if table dimensions are problematic (e.g., all zeros in a row/col)
478   if (any(rowSums(tbl) == 0) || any(colSums(tbl) == 0)) {
479     print("Error: Table contains row/column with all zeros. Cannot perform test.")
480     return(list(Test="Chi-Squared", Variable_Pair=display_name, Statistic="Error: Zero Row/Col", p_value="Error"))
481   }
482
483   # Run Chi-Squared Test with error/warning handling
484   test_result <- tryCatch({ chisq.test(tbl) },
485     warning = function(w) { print(paste("Warning:", w$message)); suppressWarnings(chisq.test(tbl)) },
486     error = function(e) { print(paste("Error:", e$message)); return(NULL) })
487
488   if (is.null(test_result)) { return(list(Test="Chi-Squared", Variable_Pair=display_name, Statistic="Error: Test Failed", p_value="Error")) }
489
490   # Check Assumptions
491   expected_counts <- test_result$expected
492   print("Expected Counts:")
493   print(round(expected_counts, 1))
494   min_expected <- min(expected_counts)
495   percent_below_5 <- mean(expected_counts < 5) * 100
496   print(paste("Minimum expected count:", round(min_expected, 2)))
497   print(paste("Percentage of cells with expected count < 5:", round(percent_below_5, 1), "%"))
498
499   assumption_met <- (percent_below_5 < 20 && min_expected >= 1)
500
501   # Prepare Results
502   test_to_report <- "Chi-Squared"
503   statistic_string <- paste0("χ²(", test_result$parameter, ") = ", round(test_result$statistic, 1))
504   p_value_formatted <- format.pval(test_result$p.value, digits = 3, eps = 0.001)
505
506   # Use Fisher's Exact Test if assumptions violated for 2x2 table
507   if (!assumption_met && nrow(tbl) == 2 && ncol(tbl) == 2) {
508     print("Chi-squared assumption violated, attempting Fisher's Exact Test...")

```



```

509     fisher_result <- tryCatch({ fisher.test(tbl) }, error = function(e) { print(paste("Fisher's test error:", e$message)); return(NULL) })
510     if(!is.null(fisher_result)){
511         test_to_report <- "Fisher's Exact"
512         statistic_string <- "N/A"
513         p_value_formatted <- format.pval(fisher_result$p.value, digits = 3, eps = 0.001)
514         print("Using Fisher's Exact Test result.")
515     } else { print("Fisher's test failed, reporting Chi-squared with caution.") }
516 } else if (!assumption_met) { print("Chi-squared assumption violated. Report with caution.") }
517
518 # Report residuals if Chi-Squared test was used and significant
519 if(test_to_report == "Chi-Squared" && !is.na(test_result$p.value) && test_result$p.value < 0.05){
520     print("Pearson Residuals (indicator for post-hoc):")
521     print(round(test_result$residuals, 2))
522 }
523 return(list(Test=test_to_report, Variable_Pair=display_name, Statistic=statistic_string, p_value=p_value_formatted))
524 }
525
526
527 # 5.2 T-Tests / Mann-Whitney U Tests Function (remains the same as previous version)
528 run_cont_vs_cat <- function(cont_var_name, cat_var_name, display_name) {
529     print(paste("--- Comparison: ", display_name, " ---"))
530     # Initial Checks
531     if (!all(c(cont_var_name, cat_var_name) %in% names(diabetes_df))) {
532         print("Error: One or both variables not found.")
533         return(list(Test="Error", Variable_Pair=display_name, Statistic="Error: Var Missing", p_value="Error"))
534     }
535     if (!is.numeric(diabetes_df[[cont_var_name]])) {
536         print(paste("Error:", cont_var_name, "is not numeric. Class:", class(diabetes_df[[cont_var_name]])))
537         return(list(Test="Error", Variable_Pair=display_name, Statistic="Error: Not Numeric", p_value="Error"))
538     }
539     if (!is.factor(diabetes_df[[cat_var_name]])) {
540         print(paste("Error:", cat_var_name, "is not a factor. Class:", class(diabetes_df[[cat_var_name]])))
541         return(list(Test="Error", Variable_Pair=display_name, Statistic="Error: Not Factor", p_value="Error"))
542     }
543
544     if (nlevels(diabetes_df[[cat_var_name]]) != 2) {
545         print(paste("Error:", cat_var_name, "does not have exactly 2 levels. Levels found:", paste(levels(diabetes_df[[cat_var_name]]),
546 collapse=" ", "(N=", nlevels(diabetes_df[[cat_var_name]]), ")")")
547         return(list(Test="Error", Variable_Pair=display_name, Statistic="Error: Not 2 Levels", p_value="Error"))
548     }
549
550     formula <- as.formula(paste(cont_var_name, "~", cat_var_name))
551
552     # Check Normality (Shapiro-Wilk - maybe unreliable for large N, supplement with plots)
553     normality_p_values <- tryCatch({
554         tapply(diabetes_df[[cont_var_name]], diabetes_df[[cat_var_name]], function(x) {
555             if (length(unique(na.omit(x))) < 3 || length(na.omit(x)) < 3) return(NA)
556             shapiro.test(x)$p.value
557         })
558     }, error=function(e){print(paste("Shapiro test failed:", e$message)); return(c(NA, NA))})
559     print("Shapiro-Wilk Normality p-values by group (NA if test failed/inapplicable):")
560     print(round(normality_p_values, 3))
561     normality_met <- all(!is.na(normality_p_values)) && all(normality_p_values > 0.05)
562
563     # Check Homogeneity of Variances (Levene's Test)
564     levene_result <- tryCatch({ leveneTest(formula, data = diabetes_df) },
565     error = function(e) { print(paste("Levene's test failed:", e$message)); return(NULL) })
566     if (is.null(levene_result)) {
567         print("Assuming unequal variances due to Levene's test error.")
568         variances_equal <- FALSE
569         levene_p <- NA
570     } else {
571         levene_p <- levene_result$Pr(>F)[1]
572         print(paste("Levene's Test p-value:", round(levene_p, 3)))
573         variances_equal <- !is.na(levene_p) && levene_p > 0.05
574     }
575
576     # Decide and Perform Test
577     test_to_report <- ""
578     statistic_string <- ""
579     p_value_formatted <- ""
580     test_result <- NULL
581
582     if (normality_met) {
583         print("Normality assumption met (based on Shapiro p > 0.05).")
584         if (variances_equal) {
585             print("Variances are equal. Performing standard t-test.")
586             test_result <- tryCatch(t.test(formula, data = diabetes_df, var.equal = TRUE), error=function(e) {print(paste("t-test error:",
587 e$message)); NULL})
588             test_to_report <- "t-Test (Equal Var)"
589         } else {
590             print("Variances are unequal. Performing Welch's t-test.")
591             test_result <- tryCatch(t.test(formula, data = diabetes_df, var.equal = FALSE), error=function(e) {print(paste("Welch t-test
592 error:", e$message)); NULL})
593             test_to_report <- "t-Test (Welch)"
594         }
595         if (!is.null(test_result)) {
596             statistic_string <- paste0("t(", round(test_result$parameter, ifelse(grepl("Welch", test_to_report), 1, 0)), ") = ",
597 round(test_result$statistic, 1))
598             p_value_formatted <- format.pval(test_result$p.value, digits = 3, eps = 0.001)
599         }
600     } else {
601         print("Normality assumption potentially violated (Shapiro p <= 0.05 or test failed). Performing Mann-Whitney U test.")
602         test_result <- tryCatch(wilcox.test(formula, data = diabetes_df), error=function(e) {print(paste("Wilcoxon test error:", e$message));
603 NULL})
604         test_to_report <- "Mann-Whitney U"
605         if (!is.null(test_result)) {
606             statistic_string <- paste0("W = ", test_result$statistic)
607             p_value_formatted <- format.pval(test_result$p.value, digits = 3, eps = 0.001)
608         }
609     }
610 }

```



```

605-   if (is.null(test_result)) {
606-     print("Statistical test failed to execute.")
607-     return(list(Test=test_to_report, Variable_Pair=display_name, Statistic="Error: Test Failed", p_value="Error"))
608-   }
609-
610-   return(list(Test=test_to_report, Variable_Pair=display_name, Statistic=statistic_string, p_value=p_value_formatted))
611- }
612-
613- # --- Add Pre-Checks before running Bivariate tests ---
614- print("--- Verifying variable status before Bivariate Analysis ---")
615- bivariate_vars <- c("Diabetes_Status", "Physical_Activity", "Family_History_of_Diabetes",
616-   "Hypertension", "Diet", "Smoking", "BMI", "Fasting_Blood_Glucose")
617- for(v in bivariate_vars){
618-   if (v %in% names(diabetes_df)) {
619-     print(paste(v, "- Class:", class(diabetes_df[[v]])))
620-     if(is.factor(diabetes_df[[v]])){
621-       print(paste(" Levels:", paste(levels(diabetes_df[[v]]), collapse=" ", "(N=", nlevels(diabetes_df[[v]]), ")"))
622-       print(" Table:")
623-       print(table(diabetes_df[[v]])) # Show counts per level
624-     } else if (is.numeric(diabetes_df[[v]])) {
625-       print(paste(" Summary:", paste(round(summary(diabetes_df[[v]]), 2), collapse=" / ")"))
626-     }
627-   } else {
628-     print(paste(v, "- COLUMN NOT FOUND in diabetes_df."))
629-   }
630- }
631- print("--- End of Variable Verification ---")
632-
633- # --- Execute Bivariate Tests ---
634-
635- # Run Chi-Squared tests
636- results_list <- c(results_list, list(run_chisq("Diabetes_Status", "Physical_Activity", "Diabetes_Status vs. Physical_Activity")))
637- results_list <- c(results_list, list(run_chisq("Diabetes_Status", "Family_History_of_Diabetes", "Diabetes_Status vs.
638- Family_History_of_Diabetes")))
639- results_list <- c(results_list, list(run_chisq("Diabetes_Status", "Hypertension", "Diabetes_Status vs. Hypertension")))
640- results_list <- c(results_list, list(run_chisq("Diabetes_Status", "Diet", "Diabetes_Status vs. Diet")))
641- results_list <- c(results_list, list(run_chisq("Diabetes_Status", "Smoking", "Diabetes_Status vs. Smoking")))
642-
643- # Run tests for continuous variables vs Diabetes_Status
644- results_list <- c(results_list, list(run_cont_vs_cat("BMI", "Diabetes_Status", "BMI by Diabetes_Status")))
645- results_list <- c(results_list, list(run_cont_vs_cat("Fasting_Blood_Glucose", "Diabetes_Status", "Fasting_Blood_Glucose by Diabetes_Status")))
646-
647- # --- Combine Bivariate Results into Table 2 ---
648- if (length(results_list) > 0) {
649-   # Filter out any potential NULL entries if a function returned NULL completely
650-   results_list_clean <- results_list[!sapply(results_list, is.null)]
651-   if (length(results_list_clean) > 0) {
652-     # Convert list of lists to data frame
653-     table2_data <- do.call(rbind.data.frame, results_list_clean)
654-     # Ensure column names are correct
655-     names(table2_data) <- c("Test", "Variable Pair", "Statistic", "p-value")
656-     # Display Table 2
657-     print("--- Bivariate Analysis Table 2 ---")
658-     print(table2_data)
659-   } else {
660-     print("Could not generate Table 2 data - all bivariate tests failed or returned NULL.")
661-   }
662- } else {
663-   print("Could not generate Table 2 data - results list is empty.")
664- }
665- }
666-
667- ...

```

## Visualization

```

# Visualizations
...{r}
# 4.3 Visualizations
# Box plot for BMI
ggplot(diabetes_df, aes(x = Diabetes_Status, y = BMI, fill = Diabetes_Status)) +
  geom_boxplot() +
  labs(title = "BMI Distribution by Diabetes Status", x = "Diabetes Status", y = "BMI (kg/m²)") +
  theme_minimal()
# ggsave("boxplot_bmi.png") # Optional save

# Box plot for Glucose
ggplot(diabetes_df, aes(x = Diabetes_Status, y = Fasting_Blood_Glucose, fill = Diabetes_Status)) +
  geom_boxplot() +
  labs(title = "Fasting Blood Glucose Distribution by Diabetes Status", x = "Diabetes Status", y = "Fasting Blood Glucose (mg/dL)") +
  theme_minimal()
# ggsave("boxplot_glucose.png") # Optional save

# Bar plot for Family History
ggplot(diabetes_df, aes(x = Family_History_of_Diabetes, fill = Diabetes_Status)) +
  geom_bar(position = "fill") + # Use position="fill" for proportions
  labs(title = "Proportion with Diabetes by Family History", x = "Family History of Diabetes", y = "Proportion", fill = "Diabetes Status") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal()
...

```

